

Philopateer Boctor

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EDUCATION

Yale University

New Haven, CT

Bachelor of Science in Mechanical Engineering (ABET). GPA: 3.5

Aug. 2022 – May 2026

Relevant Coursework: Mechanical Design; Machine Elements; Strength of Materials; Solid Mechanics Lab; Dynamics; Thermodynamics; Fluid Mechanics; Soft Robotics; Computer-Aided Engineering; Materials Science, Mechatronics

EXPERIENCE

Verustruct

June 2025 – Aug 2025

Mechanical Engineering Intern

New Haven, CT

- Owned the mechanical architecture for both the horizontal and vertical retraction/extension mechanisms of Verustruct's multi-directional Translational Slip-Form (TSF) system, developing the first integrated, full-scale proof-of-concept adopted for the metal prototype now being manufactured at the HAX accelerator.
- Evaluated mechanism layouts, packaging tradeoffs, and manufacturable geometries in Siemens NX, creating a compact slipform envelope that met two-person lift constraints while preserving internal volume for pumps, sensors, and future robotic subsystems.
- Improved the slipform's retraction reliability through prototype testing and failure analysis, identifying binding events under actuator load and adjusting member stiffness, clearances, and assembly geometry to allow the mechanism to cycle smoothly.
- Led weekly design reviews with the founder using CAD models, prototype observations, and manufacturability tradeoffs to guide decisions on final geometry, assembly strategy, and integration envelope for the company's flagship construction robot.

Yale University GRAB Lab

Oct. 2023 – May 2025

Undergraduate Researcher | PI: Prof. Aaron Dollar

New Haven, CT

- Improved the GRAB Lab's underactuated Stewart Hand by replacing its pulley-based 1-to-3 differential with a compact gear-driven design, using mechanism iteration to improve planar positioning accuracy and create more consistent torque distribution to all three fingers.
- Engineered a complete multi-stage drivetrain in SolidWorks, iterating custom miter gears through tolerance analysis, 3D printing, and benchtop testing until a single motor input produced coordinated grasping motion across all three fingers.

PROJECTS

Translational Slipform Feed System | Capstone Project

Aug 2025 – Apr 2026

- Led a four-person capstone team in developing a material delivery and extrusion test platform for Verustruct's translational slipform robot, designing a system to condition pumpable material for controlled deposition.
- Designed a conveyor-driven test platform in SolidWorks to mimic robot motion, enabling repeatable extrusion testing under controlled process conditions.
- Integrated LabJack DAQ sensing for temperature, humidity, VMA flow rate, and conveyor speed to support quantitative testing of material behavior, process parameters, and system performance.
- Completed final prototype testing and identified rapid aggregate segregation as a key failure mode, leading to continuous in-line mixing recommendations and informing the final system design.

In-Pipe Soft Robot for Locomotion

Jan 2025 – May 2025

- Led a team of graduate students to design and fabricate a McKibben-actuated soft crawler for in-pipe locomotion, translating research literature into mold-ready CAD, custom silicone actuators, and a working prototype.
- Iterated mold geometry, silicone casting methods, actuator alignment, and wall-thickness control to reduce rupture risk and enable successful straight-pipe locomotion.

TECHNICAL SKILLS

Design & Analysis: Mechanical Design, Mechanism Design, 3D Modeling, Rapid Prototyping, Tolerance Analysis, Prototype Testing, Failure Analysis, FEA, GD&T

Software & Programming: SolidWorks, Siemens NX, Onshape, MATLAB, Intelli-MAX, Arduino, L^AT_EX

Machining & Fabrication: FDM/SLA 3D Printing, Lathe, Mill, Waterjet, Laser Cutter, Common Machine Shop Tools